



HWA CHONG INSTITUTION
JC2 Preliminary Examinations
Higher 3

MATHEMATICS

9824/01

Paper 1

22 September 2016

3 hours

Additional Materials: Answer Paper
List of Formulae (MF15)

READ THESE INSTRUCTIONS FIRST

Write your name and CT group on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams or graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** the questions.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use a graphic calculator.

Unsupported answers from a graphic calculator are allowed unless a question specifically states otherwise.

Where unsupported answers from a graphic calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

You are reminded of the need for clear presentation in your answers.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

This document consists of 7 printed pages.



HWA CHONG INSTITUTION
2016 C2 Preliminary Examinations
Higher 3

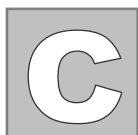
MATHEMATICS

9824/01
22 September 2016
3 hours

Name:

CT:

--	--	--	--	--



OVER PAGE

1. Write your name, CT group and calculator model(s) in the spaces provided.
2. Arrange your answers in numerical order.
3. Place this cover page on top of your answer sheets and fasten them securely together with the string provided.

For Examiner's Use

Question No.	Marks Obtained	Total Marks	Remarks
1		13	
2		14	
3		13	
4		10	
5		7	
6		6	
7		12	
8		13	
9		8	
10		4	
TOTAL		100	

Graphing Calculator Model:

Scientific Calculator Model:

- 1 (a) Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function such that

$$f(x) = 5 + 3 \int_0^x f(t) dt,$$

where $x > 0$.

- (i) Find $f(0)$. [1]

- (ii) Express $f'(x)$ in terms of $f(x)$. [1]

- (iii) By considering the derivative of $e^{-3x}f(x)$, find $f(x)$. [4]

- (b) (i) By considering the application of the AM-GM Inequality on the numbers $1, 3, 5, \dots, 2n-1$, prove that $n^n \geq 1 \cdot 3 \cdot 5 \cdot \dots \cdot 2n-1$ for all $n \in \mathbb{R}^+$. [3]

- (ii) Hence prove that $(n^2 + n)^n \geq (2n)!$. [4]

- 2 (a) Find the sum of $\tan x \tan 2x + \tan 2x \tan 3x + \dots + \tan Nx \tan (N+1)x$ for $x \neq 0$. [4]

- (b) The terms of a sequence $\{a_n\}$ are given by

$$a_n = \begin{cases} \frac{n+1}{2}, & n = 2k-1 \text{ where } k \in \mathbb{R}^+, \\ 2^{\frac{n}{2}}, & n = 2k \text{ where } k \in \mathbb{R}^+. \end{cases}$$

Let $b_n = \frac{a_{2n-1}}{a_{2n}}$ and $S_n = b_1 + b_2 + \dots + b_n$.

- (i) By considering $S_n - \frac{1}{2}S_n$, show that $S_n = 2 - (n+2)\left(\frac{1}{2}\right)^n$. [5]

- (ii) Hence prove by mathematical induction that $2 - S_n < \frac{1}{n}$ for all $n \geq 6, n \in \mathbb{R}^+$. [5]

- 3 (i) Let $f : [a, b] \rightarrow [a, b]$ be a continuous function where $a, b \in \mathbb{R}^+$ and $a < b$. Prove that there exists a solution for the equation $f(x) = x$. [3]

- (ii) It is given that there exists a constant $0 \leq \lambda < 1$ for which f from part (i) satisfies the inequality

$$|f(x_2) - f(x_1)| \leq \lambda |x_2 - x_1|$$

for all $x_1, x_2 \in [a, b]$. Prove that the solution for the equation $f(x) = x$ is unique. (Hint: If $f(x) = x$ has a unique solution, then if α and β are a pair of roots of $f(x) = x$, then $\alpha = \beta$.) [4]

- (iii) Let $\{x_n\}$ be a sequence such that $x_1 = a$, $x_n = f(x_{n-1})$ and $n \in \mathbb{R}^+$. By using the result in part (ii), prove that $\lim_{n \rightarrow \infty} x_n = \alpha$. [6]

- 4 A particular solution of the differential equation

$$(2x^2 + x - 1) \frac{dy}{dx} + (10x + 4)y = \frac{1}{2x^2 + x - 1}$$

has $y = 0$ when $x = 1$.

- (i) Use the Euler method with step size 0.5 to estimate y at $x = 3$. [3]

- (ii) By considering $\frac{d}{dx} \left[\ln \left\{ (x+1)^2 (2x-1)^3 \right\} \right]$, solve the differential equation given that $y = 0$ when $x = 1$. [4]

- (iii) Using the result in part (ii), find the exact value of $y(3)$ and hence compute the percentage error in part (i). [2]

- (iv) Suggest one way to improve the accuracy of the numerical method used in part (i). [1]

- 5 When some molten raw metals are mixed together, the rate at which the new alloy is formed is modeled by the differential equation

$$\frac{dL}{dt} = p(\alpha - L)(\beta - L),$$

where L is the mass (in grams) of the new alloy formed. It is given that $p > 0$, where p is the constant of proportionality, and $0 < \alpha \leq \beta$.

- (a) Given that $\alpha < L < \beta$ when $t = 0$, sketch a phase line diagram and predict the behaviour of L as $t \rightarrow \infty$. [2]
- (b) Consider the case when $\alpha = \beta = 2$.
- (i) Sketch a phase line diagram to predict the behavior of L as $t \rightarrow \infty$ when $0 < L(0) < \alpha$. [2]
- (ii) Solve the differential equation for the case when $p = 1$. Hence sketch the particular solution satisfying $L(0) = 1$. [3]

- 6 The population y (in kilotons) of the Longfin Inshore squid with per capital growth rate 6% is subjected to harvesting at a rate H kilotons per year. The harvested population is modeled by the differential equation (Schaefer model)

$$\frac{dy}{dt} = k \left(1 - \frac{y}{N} \right) y - H,$$

where N is the carrying capacity and H is the rate of harvesting. It was estimated that there were 5500 (in kilotons) of the squid in year 2000 and the carrying capacity is 12000 (in kilotons).

- (a) Calculate the maximum sustainable number of squids (in kilotons) that could be caught in year 2000. [4]
- (b) Calculate the maximum sustainable number of squids (in kilotons) that could be caught per year eventually. [2]

- 7 (a) Solve the equation $\frac{d^2\theta}{dt^2} + \lambda \frac{d\theta}{dt} + \mu\theta = 0$, where λ and μ are constants such that $4\mu > \lambda^2$. [3]

- (b) (i) By using the substitution $x = \ln t$, show that the differential equation $t^2 \frac{d^2y}{dt^2} + t \frac{dy}{dt} - y = 0$ can be reduced to $\frac{d^2y}{dx^2} - y = 0$. [4]

- (ii) Mr Lee bought an insurance policy with an annual premium of \$2800. He pays the premium at the start of each new year. The surrender value, $\$(100y)$ is the amount of money he receives if he terminates his policy in the t^{th} year. The variables y and t are related by the differential equation

$$t^2 \frac{d^2y}{dt^2} + t \frac{dy}{dt} = y.$$

Given that $y = 0$ and $\frac{dy}{dt} = 60$ when $t = 1$, find y in terms of t . Find the amount of profit he would make if he terminates his policy in its 5th year.

[5]

- 8 (a) (i) By the method of listing, find the number of integer solutions to $x_1 + x_2 + x_3 = 30$, where $0 \leq x_1 \leq 10$, $0 \leq x_2 \leq 11$ and $0 \leq x_3 \leq 11$. [1]

- (ii) By viewing part (i) as a distribution problem, verify the answer to part (i) using the inclusion-exclusion principle. [4]

- (b) For $0 \leq x_1 \leq n_1$, $0 \leq x_2 \leq n_2$ and $0 \leq x_3 \leq n_3$, write down the answer to part (i) when $n_1 + n_2 + n_3 < 30$. [1]

- (c) Find the number of integer solutions to $x_1 + x_2 + x_3 \leq n$, where $0 \leq x_1 \leq 2$, $0 \leq x_2 \leq 1$ and $0 \leq x_3 \leq n - 2$ for some integer $n \geq 2$. [2]

- (d) Let $\{x_1, x_2, \dots, x_n\}$ be a set of integers. Show that there exists a subset $\{x_k, x_{k+1}, \dots, x_{k+m}\}$ whose sum $x_k + x_{k+1} + \dots + x_{k+m}$ is divisible by n . [5]

- 9 In an $n \times n$ grid, we draw rectilinear paths from $(0,0)$ to (n,n) , going in only positive x and y directions as shown in the following example:

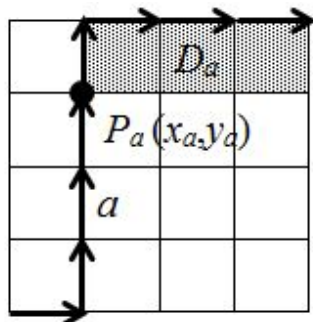


Figure 1: Example of a path in a 4×4 grid.

For a given path a , the point $P_a(x_a, y_a)$ denotes the point where a first touches $y = x + 2$ and the region D_a contains all the points (x, y) that satisfy $x \geq x_a$ and $y \geq y_a$.

- (i) Show that when D_a is reflected in the line $y = x + 2$, the resultant path lies in a $(n-2) \times (n+2)$ grid. [3]
- (ii) Hence by considering the bijection principle, or otherwise, show that the number of paths that stay on or below the line $y = x + 1$ is given by $\frac{4n+2}{(n+1)(n+2)} \binom{2n}{n}$. [5]

- 10 In computer science, *Merge Sort* is a well-known sorting algorithm. One of the simpler versions of Merge Sort works by taking an input of n distinct positive integers and outputs these same numbers in ascending order, where n is a power of 2. If n is 2^0 , then Merge Sort does nothing, for the list is already sorted. If n is more than 1, then the first and second halves of the list are each sorted recursively by Merge Sort. The two halves are then sorted again by Merge Sort to form a final sorted list.

For example, suppose we have the list $\{3,1,5,4\}$. Since n is more than 1, the first and second halves are each sorted recursively by Merge Sort into $\{1,3\}$ and $\{4,5\}$. Merge Sort is then performed again to form $\{1,3,4,5\}$.

Merge Sort merges two halves by comparing the leading terms of the two halves and emitting the smaller number. If one half is empty, the whole of the other half is emitted. There is no comparison when one half is empty.

The following table illustrates these processes:

Step	First Half	Second Half	Output
1. Split $\{3,1,5,4\}$ into two halves.	$\{3,1\}$	$\{5,4\}$	None
2. Split $\{3,1\}$ into two halves.	$\{3\}$	$\{1\}$	None
3. <u>Comparison</u> .	$\{\underline{3}\}$	$\{\underline{1}\}$	$\{1\}$
4. Emit the other half.	$\{3\}$		$\{1,3\}$
5. Split $\{5,4\}$ into two halves.	$\{5\}$	$\{4\}$	None
6. <u>Comparison</u>	$\{\underline{5}\}$	$\{\underline{4}\}$	$\{4\}$
7. Emit the other half.	$\{5\}$		$\{4,5\}$
8. Gather the two sorted halves.	$\{1,3\}$	$\{4,5\}$	None
9. <u>Comparison</u>	$\{\underline{1},3\}$	$\{\underline{4},5\}$	$\{1\}$
10. <u>Comparison</u>	$\{\underline{3}\}$	$\{\underline{4},5\}$	$\{1,3\}$
11. Emit the other half.		$\{4,5\}$	$\{1,3,4,5\}$

Hence for this example, there are 4 comparisons.

- (i) Let a_n be the maximum number of comparisons required for a list of n numbers, where n is a power of 2. Find the values of a_2 and a_4 . [1]
- (ii) Find a recurrence relation for a_n , for $n \geq 2$ and n is a power of 2. Justify your answer. [3]